

**Table 4. Relative performance on ACIC and IHDP.** We report relative AUTO C, overall win rate, and the *Rank-Learner* win rate. Relative AUTO C is computed by min-max scaling AUTO C within each setting. Overall win rate is the fraction of settings in which a method achieves the highest AUTO C. *Rank-Learner* win rate is the fraction of settings in which *Rank-Learner* achieves a higher AUTO C than the corresponding method.

| Method              | ACIC            |                  |                              | IHDP            |                  |                              |
|---------------------|-----------------|------------------|------------------------------|-----------------|------------------|------------------------------|
|                     | Relative AUTO C | Overall win rate | <i>Rank-Learner</i> win rate | Relative AUTO C | Overall win rate | <i>Rank-Learner</i> win rate |
| T-learner           | 44.4%           | 12.5%            | 75.0%                        | 22.5%           | 0.0%             | 82.0%                        |
| DR-learner          | 46.0%           | 12.5%            | 87.5%                        | 61.2%           | <b>42.0%</b>     | 54.0%                        |
| Plug-in ranker      | 55.1%           | 12.5%            | 62.5%                        | 49.6%           | 16.0%            | 72.0%                        |
| Rank-Learner (ours) | <b>84.7%</b>    | <b>62.5%</b>     | —                            | <b>69.2%</b>    | <b>42.0%</b>     | —                            |